



A Qualitative Enquiry of Hypoglycemia and the Social Determinants of Health: The InHypo-DM Study, Canada

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Introduction: A significant body of research exists on the impact of the social determinants of health (SDoHs) on diabetes care and general health outcomes. However, less is known about health care practitioners' (HCPs') perspectives and experiences regarding the impact of the SDoHs on their patients with Type I and Type 2 diabetes mellitus and how this affects the prevention and treatment of hypoglycemia. **Method:** A descriptive qualitative study, derived from the InHypo-DM (Canada) research program. A purposive sampling technique was used to recruit participants residing in southwestern Ontario, Canada, for a 30- to 45-min semistructured interview. Individual and team analysis of interviews was conducted to identify overarching and subthemes. Twenty HCP participants, including endocrinologists, family physicians, and allied health care practitioners, were recruited. Seven were Certified Diabetes Educators. **Results:** Participants articulated 2 overarching components of the SDoHs: patients' socioeconomic issues and psychosocial issues. They highlighted two socioeconomic issues: occupation type and poverty. Participants also emphasized 3 areas pertaining to patients' psychosocial issues: stage in the life cycle (e.g., elderly), social isolation, and mental health. **Discussion:** This study emphasizes the need for conducting detailed and comprehensive social histories during clinical diabetes assessments, as well as the necessity of adequate clinical time and resources for HCPs and patients to address these issues in the context of hypoglycemia management.

Keywords: hypoglycemia, diabetes mellitus, social determinants of health, health care practitioners, qualitative research

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Evidence suggests that traditional approaches to glycemic management that focus solely on blood glucose targets are insufficient at reducing rates of hypoglycemia (Goodall & Halford, 1991; Peyrot et al., 2005). Rather, attention should be expanded toward elucidating potential situational, environmental, and psychosocial factors that may impact the ability of the person living with Type 1 diabetes mellitus (T1DM) or Type 2 diabetes mellitus (T2DM) to manage hypoglycemia. This knowledge gap is compounded by a lack of understanding of the role these factors have on health care practitioners (HCPs) and their ability to care for patients with diabetes (Williamson et al., 2014). With hypoglycemia being one of the most common and serious diabetes-related adverse events, it is imperative to understand the factors contributing to this problem in context of caring for the whole person.

Typically, hypoglycemia is described in three different categories: nonsevere, severe, and nocturnal. In nonsevere cases, affected individuals may experience symptoms of sweating, trembling, and hunger but are still neurologically aware enough to be able to treat themselves with a fast-acting sugar. In severe situations, the person's blood sugar may be so low as to require third-party assistance to treat as this person may be confused or have lost consciousness. Nocturnal hypoglycemia is especially concerning as the individual may be asleep and therefore not aware of early nonsevere symptoms and slip into a state of severe hypoglycemia (Cheng & the Canadian Diabetes Association Clinical Practice Guidelines Expert Committee, 2013). Canadian data using a convenience sample of 202 and 133 people with T1DM and T2DM, respectively, treated with insulin reported the rate of severe hypoglycemia as 2.6 and 5.9 events per person-year in people with T1DM and T2DM, respectively (Leiter et al., 2005). More recently, the "understanding the impact of *HYPO*glycemia on *Diabetes Management* study" or InHypo-DM (Canada) study was launched to improve our understanding of the frequency of hypoglycemia (nonsevere, nocturnal, and severe) in patients with T1DM and T2DM at risk for hypoglycemia. Data collected via self-reported nationwide survey of 552 adults with diabetes (17% T1DM, 82.9% T2DM) treated with insulin and/or sulfonylurea reported that the population-based burden of

hypoglycemia among people with diabetes is substantial among people with T1DM (2.5 per person-year) and T2DM (2.4 per person-year; Harris et al., 2016).

From HCPs' perspective, hypoglycemia is often cited as a major barrier to achieving optimal glycemic targets as it can limit the ability to intensify treatment regimens (Kelly et al., 2009). To understand the challenges HCPs face while managing diabetes, the DAWN2 study investigated a total of 4,785 HCPs caring for people with diabetes across 17 countries (Holt et al., 2013). Their online survey explored various HCPs' perspectives, including the level of diabetes training, perceived support from other HCPs in helping to manage patients with diabetes mellitus (DM), and perceptions of the burden of diabetes on their patients. Over half of the participants indicated that they had discussed emotional issues with their patients yet were frustrated by the lack of resources or support to then address these emotional issues in a productive manner (Holt et al., 2013).

If HCPs understand the role that patient-related social determinants of health (SDoHs) may have on their patients' ability to manage diabetes, a reduction in hypoglycemia rates may follow suit. Two recent reviews (Clark & Utz, 2014; Walker, Smalls, Campbell, Strom Williams, & Egede, 2014) present a lens through which we can study the complex factors of SDoHs. To understand what factors are considered SDoHs, the authors of these studies described five domains: (1) health and health care, (2) neighborhood and built environment, (3) economic stability, (4) education, and (5) social and community context. Building on the work of Brown et al., 2004; Walker, Gebregziabher, Martin-Harris, & Egede, 2014b described two overarching dimensions of SDoHs: (1) socioeconomic (e.g., education, employment, income) and (2) psychosocial (e.g., psychological distress, perceived stress, social support).

Economic stability or socioeconomic status (SES) has been extensively studied as a SDoH that influences the prevalence of diabetes (Connolly, Unwin, Sherriff, Bilous, & Kelly, 2000; Evans, Newton, Ruta, MacDonald, & Morris, 2000) and its role on glycemic control (Borschuk & Everhart, 2015; Khanolkar et al., 2016; Walker, Gebregziabher, et al., 2014b; Walker, Smalls, et al., 2014; Walker, Smalls, & Egede, 2015). Lower SES is associated with

increased DM prevalence and poor glycemic control. Specifically, SES has been identified as a risk factor for severe hypoglycemia (Leese et al., 2003; Sawka et al., 2007), and as Seligman, Davis, Schillinger, & Wolf, 2010 described in a population with diabetes and low income, this may be in part related to food insecurity. Lower employment rates have also been found to contribute to the incidence of T2DM (Kumari, Head, & Marmot, 2004). Finally, financial insecurity has been linked to cost-related medication underuse, which can increase patients' risk for serious complications (Berkowitz et al., 2015; Piette, Heisler, & Wagner, 2004; Piette, Wagner, Potter, & Schillinger, 2004).

The link between psychosocial issues as a SDoH, specifically depression and glycemic control, is well established (Ali, Stone, Peters, Davies, & Khunti, 2006; Barnard, Skinner, & Peveler, 2006; Chiu, Wray, Beverly, & Dominic, 2010; de Groot, Anderson, Freedland, Clouse, & Lustman, 2001; Fisher et al., 2008). SDoHs that are more specific to psychosocial issues include social support or isolation, perceived stress, and mental illness, all of which have been linked to glycemic control (Akhter, Turnbull, & Simmons, 2016; Holt et al., 2016; Joensen, Almdal, & Willaing, 2013; Smalls, Gregory, Zoller, & Egede, 2015a; Walker et al., 2015; Walker, Gebregziabher, Martin-Harris, & Egede, 2014a).

To date, only one qualitative study has examined the impact of SDoHs on patients with diabetes (Akhter et al., 2016). The literature also reveals that the SDoHs are frequently only studied in patients with T2DM (Connolly et al., 2000; Kumari et al., 2004; Smalls, Gregory, Zoller, & Egede, 2015b; Walker et al., 2015; Walker, Gebregziabher, et al., 2014a), with few

studies capturing the effect of the SDoHs on glycemic control among adults with T1DM.

The InHypo-DM study was designed to not only capture a better understanding of the frequency of hypoglycemia in T1DM and T2DM but also to collect information regarding the complex determinant factors of hypoglycemia management behavior among people with diabetes, significant others of people with diabetes, and the HCPs involved in their care. The present analysis draws specifically upon a subset of the data, namely, the qualitative findings of the InHypo-DM study. This subset analysis (as a part of the larger overall InHypo-DM study) revealed a theme of the importance of SDoHs. This report explores HCPs' perceptions of the role SDoHs have on their patients' ability to prevent and treat hypoglycemia.

Method

This study used a descriptive qualitative approach, which is well suited to the exploratory nature of our investigation (Sandelowski, 2000). Ethics approval for this study was received from the University of Western Ontario's Research Ethics Board (REB 105992).

Participant Recruitment

Participants were recruited from southwestern Ontario, Canada, an area in which the local health integration network has reported a diabetes prevalence of approximately 8.8% (Booth et al., 2012). The HCPs practiced in a variety of settings, including hospital, specialist, primary care, and community locations. Twenty care practitioners were recruited (see Table 1). Within this purposive sample, maximum varia-

Table 1
Participant Demographics

Participant type	Number	Male	Female	Mean age (y)	No. of years in practice, mean	No. of CDEs per participant type
Endocrinologist	4	1	3	51	18	—
Family physician	4	2	2	46	15	—
Nurse practitioner	3	0	3	55	22	1
Registered nurse	3	0	3	44	22	2
Registered dietitian	2	0	2	53	28	2
Pharmacist	4	3	1	49	23	2

Note. CDE = Certified Diabetes Educator.

tion sampling (professional and practice settings) was used.

Data Collection

Upon gaining consent, a 30- to 45-min semi-structured interview was conducted with each participant. Questions from the interviews relevant to this study were as follows: “What do you think are some of the barriers to patients regarding managing their hypoglycemia?” and “What patient characteristics affect their ability to manage their hypoglycemia?” We did not specifically ask about the impact of SDoHs on hypoglycemia. The interviews, conducted by members of the research team in a location convenient to the participants, were audiotaped and transcribed verbatim. Data collection ceased upon reaching saturation in that there were no new emerging themes.

Data Analysis

The data analysis was both iterative and interpretative using both individual and team analysis. In the initial phase of the analysis, three members of the research team independently reviewed each transcript to identify key concepts emerging from the data. We then met to compare our independent reviews, culminating in the development of the coding template, which evolved over the course of the analysis. NVivo 10 software was used for coding and organizing the data.

Upon completion of this phase, we then met to further synthesize and interpret the main themes using the techniques of immersion and crystallization (Borkan, 1999). Immersion involves researchers’ complete engagement in the data, therefore sensitizing them to the tone, range, mood, and context of the data during the analysis (Borkan, 1999). Crystallization reflects the ultimate synthesis of the main themes, as expressed by participants.

Throughout the analysis, the research team identified exemplary quotes reflecting the main themes specific to this study. We coded participants’ key phrases and ideas under a variety of codes, such as poverty, mental health, job security, and social support, resulting in 30 separate codes. To synthesize these numerous codes further, two of the researchers reexamined the data. As Walker, Gebregziabher, et al. (2014b) outlined, the validated

SDoH conceptual framework consisting of the two main components of (1) socioeconomic and (2) psychosocial was used for this next level of analysis of how SDoHs may impact diabetes outcomes.

Credibility and Trustworthiness

Credibility and trustworthiness of the data were enhanced through verbatim transcripts, field notes generated after each interview, and independent and team analysis. Reflexivity was essential during the analysis. As we came from different backgrounds (e.g., social work, family medicine), each of us reflected on how our values and experiences shaped the interpretation and reporting of the data. As a research team, we reflected and discussed how our values and experiences could shape the interpretation and reporting of the data.

Results

Similar to the framework described by Walker, Gebregziabher, et al. (2014b), the HCPs participating in this study described two overarching components of the SDoHs: socioeconomic and psychosocial issues.

Socioeconomic Issues

From the HCP participants’ perspective, they described two specific areas pertaining to patients’ socioeconomic issues: (1) occupation type, including the impact of shift work and the number of workdays missed due to hypoglycemia, and (2) poverty, including the inability to purchase test strips.

Participants described their belief that certain occupations made it very challenging for patients to manage their hypoglycemia.

There’s some occupations that your heart sinks when they come in. So people who cook for a living . . . people who drive long-haul trucks . . . certain types of occupations are unforgiving when it comes to the demands of being on a glycemic regimen. (NP-1)

For patients whose occupations required shift work, managing their hypoglycemia could be more difficult. “Shift workers. I have a few patients who work almost exclusively at night, and we have found that challenging” (FP-2). The participants noted loss of days at work for patients with hypoglycemic events: “Patients

who have many hypoglycemia miss days of work” (Endo-3). Furthermore, participants expressed concern for patients whose absenteeism could jeopardize their employment security. “Patients could be at risk of losing their employment if they have repeated episodes of hypoglycemia” (RNCDE-1).

Participants also described how poverty was a powerful influence on patients’ management of their hypoglycemia and could result in, for example, “lack of consistency regarding where they get their medications or how they come into the clinic. So no regular follow-up” (FP-1). Participants also noted how a lack of finances prohibited patients’ purchase of test strips. “Probably the biggest barrier is financial regarding having the money to test their blood sugar regularly” (RNCDE-2). The consequences were significant as patients were unable to routinely monitor their blood sugars and make necessary adjustments. “If they cannot afford the strips . . . they may be trying to save their strips; they run out of strips so they’re not getting the feedback of what’s happening with their sugars” (NP-2).

Participants expressed concern when financial constraints prevented patients from purchasing necessary medications, often resulting in unsafe medication use.

What I still find incredibly frustrating is the person who needs medication but can’t afford them and so they manipulate their medicines the best they can. It’s an economic issue . . . and they manipulate them in a way that’s dangerous because they can’t afford medicines. (NP-1)

Even if patients had coverage under a government drug plan, there were certain restrictions: “It pays for the strips but it doesn’t pay for the lancing device. . . . So there are components of the system that can be the tipping point for some patients” (NP-1).

Psychosocial Issues

Participants emphasized that from their perspectives, there were three specific areas pertaining to patients’ psychosocial issues: (1) stage in the life cycle, (2) social isolation, and (3) mental health. A patient’s position in the life cycle could influence his or her ability to appropriately manage their hypoglycemia. “I guess depending on what stage in your life you’re at the importance of your health is different. If you’re a child your parents are going

to be a potential help or barrier” (RN-1). Participants who cared for new mothers with diabetes had concerns about their ability to juggle the challenges of parenthood and hypoglycemia management. “I work a lot with new mothers, so if the baby is crying they might think it’s more important to deal with the baby than testing their blood sugar to make sure they’re safe” (RD-2).

A specific subcohort monitored closely by participants was the elderly. “I constantly worry about our elderly . . . they have other comorbidities, and they also do not feel their hypoglycemia” (FP-1). Of particular concern were elderly patients who lived alone. “We have a hard time sending someone home on insulin if they are an old lady and they are living alone. We would look at other options before just because we know the concerns around that” (RD-1). Participants perceived elderly patients as being at risk of social isolation with a diminished quality of life.

I had a patient who just recently retired and was having multiple low blood sugars and it was really impacting her ability to be social. So she wasn’t going out to things. She planned them and then she’d have a low blood sugar and then she felt so horrible, and it took her so long to recover that she didn’t feel up to going out and participating in social gatherings. So I think quality of life is really impacted when somebody’s struggling. (RNCDE-1)

Social isolation, coupled with the absence of family involvement, was also viewed by participants as particularly challenging for patients lacking both instrumental and emotional support of maintaining glycemic control. “I think family if you don’t have someone to help you and encourage you that’s a huge barrier” (FP-4). Helping patients who were housebound manage their hypoglycemia was similarly a concern. “Patients who are housebound. So over the phone, it’s a bit tough to talk about that [hypoglycemia] sometimes” (Pharm-5).

Participants explained how patients’ mental health issues could present obstacles to care. “Mental illness where their coping mechanisms are poor or significant mental illness where they’re unable to take care of themselves” (FP-1). Furthermore, some participants described how they could be distracted from addressing hypoglycemia management due to other issues impinging on the patient’s life.

So, for example, depression or something else going on in their life [then] you might not discuss hypoglycemia because you are distracted by the other major issues.

If someone comes in and they just got kicked out of their apartment or something else major is going on, that would prevent me from managing or discussing the hypoglycemia. (RD-2)

Discussion

The findings of this study, part of the larger InHypo-DM Canada study (Harris et al., 2016), emphasized the importance of HCPs to understand the SDoHs that may, in turn, affect their patients' ability to manage hypoglycemia. The two main overarching dimensions of SDoHs highlighted by our participants parallel the work of Walker, Gebregziabher, et al. (2014b): socioeconomic issues and psychosocial issues. However, our findings differ from their work in that our participants highlighted different components within these two dimensions as key for (1) socioeconomic issues (occupation, including shift work and poverty) and (2) psychosocial issues (stage of life cycle, social isolation, and mental health).

Occupational status has long been established as a major SDoH (Marmot et al., 1991). SES has been identified as a risk factor for severe hypoglycemia (Leese et al., 2003; Sawka et al., 2007). However, our findings relating occupation to hypoglycemia management risk mark a novel finding for the SDoH issue. Our participants, in discussing hypoglycemia management, honed in on certain occupational roles (e.g., cook, long-haul truck driver), which could potentially make hypoglycemia management more difficult. For certain occupational roles, participants emphasized that some employment conditions such as shift work may be an area of consideration when assisting their patients in avoiding a hypoglycemia event. This supports previous InHypo-DM findings that showed that shift work was significantly associated with both nocturnal and severe hypoglycemia (Harris et al., 2016). Finally, our participants also acknowledged how frequent hypoglycemia events occurring at the workplace could place their patient's employment status in jeopardy.

Living in poverty, another well-established SDoH, was also noted by our participants. From their perspective, financial insecurity could lead to cost-related medication misuse

as well as underuse. While the latter is well documented in the literature (Piette, Heisler, et al., 2004; Piette, Heisler, Horne, & Caleb Alexander, 2006; Piette, Wagner, et al., 2004), our participants described a misuse of medications as unsafe and potentially contributing to a hypoglycemic event. Canada's universal public health care does not include prescription drugs. Prior research has shown that patients with diabetes who have drug coverage demonstrated less cost-related medication underuse (Piette, Wagner, et al., 2004); in contrast, patients with cost-related poor medication adherence have poorer diabetes control and higher emergency department utilization (Berkowitz et al., 2015).

In addition, lack of communication between patients and practitioners about cost-related medication use has been attributed to insufficient time to discuss these issues and the perceived belief that there was no solution to the patients' financial problems (Alexander, Casalino, Tseng, McFadden, & Meltzer, 2004). The inability to obtain necessary items such as test strips or medications could be life-threatening and must be addressed during clinical encounters. With the expansion of interprofessional teams engaged in the delivery of primary care, other disciplines such as social workers can assist family physicians in guiding patients to resources addressing cost-related medication use (Brown, Ryan, & Thorpe, 2016; Rabovsky et al., 2017).

Our study findings confirm how a patient's stage in the life cycle will influence their response to the illness experience (Stewart et al., 2014). Of particular concern to the participants were their elderly patients' risk of social isolation and subsequent risk of hypoglycemia. Attention to the specific needs of elderly patients with diabetes has been advocated in prior research as this population may be more vulnerable due to social isolation, comorbidity, and financial stresses (DiMatteo, 2004; Gunn, Seers, Posner, & Coates, 2012; Hempler, Joensen, & Willaing, 2016; Holt et al., 2016; Zhang, Norris, Gregg, & Beckles, 2007).

Mental health issues, as noted by participants, could seriously impede patients' abilities to optimally manage their hypoglycemia. The impact of mental health on hypoglycemia management existed not only for patients with formal psychiatric diagnoses but also for patients

experiencing chronic stress, for example, relationship issues. The role of chronic stress has been reported previously in relation to glycemic control (Akhter et al., 2016; Hill, Nielsen, & Fox, 2013; Liburd, Jack, Williams, & Tucker, 2005; Maddigan, Feeny, Majumdar, Farris, & Johnson, 2006; Raphael et al., 2003; Walker, Gebregziabher, et al., 2014a). Furthermore, the DAWN2 study echoes these important findings, revealing that patients who were identified to have worse psychological outcomes were at higher risk to experience complications from diabetes and have a higher incidence of hypoglycemia (Nicolucci et al., 2016). Finally, our participants described how patients' mental health issues could serve as a distraction limiting their attention to the patients' hypoglycemia management. Similar situations have been reported by family physicians while attending to the physical needs of their patients with serious mental illness (Brown, Lent, Stirling, Takhar, & Bishop, 2002).

While there appears to be a willingness to engage patients with diabetes in conversations about their emotional problems, HCPs in the DAWN2 studies have previously reported a lack of resources to assist people with diabetes who are at risk for depression or experiencing psychological distress (Holt et al., 2013). Addressing the complex and debilitating SDoHs some patients may be experiencing can be daunting for HCPs, potentially being further hampered by a lack of resources.

The recently published article by Dowell et al. (2018) offers practical solutions as to how to better address social issues during diabetes clinical care stemming from their qualitative research. They found that while HCPs employed a high quality of communication skill with their patients, it was often focused on the biomedical aspects of diabetes and less so on the social context or the reality of the patient, that is, ignoring content knowledge a patient may already possess about diabetes. They suggested that these issues could be mitigated if a patient-centered approach is taken when obtaining patient histories and finalizing management plans. Expanding on the concept of patient-centered care, it is also important to take into consideration both the patient's proximal context (e.g., family supports) and distal context (e.g., socioeconomic status) when exploring the SDoHs in relation to hy-

poglycemia and diabetes (Stewart et al., 2014). As in previous studies, the HCPs of the Dowell et al. study also cited lack of time. However, upon further analysis, it was discovered that educational topics were often repeated between different HCPs. Avoiding duplication of teaching within the diabetes team may allow for more time to focus on social topics.

Our findings strongly suggest the need for a detailed and comprehensive social history on persons with diabetes that resonates with patient-centered care. Behforouz, Drain, and Rhatigan (2014) call for a more comprehensive approach to obtaining a social history that would include, for example, asking about employment (work hours, stresses), food security, medication access and affordability, social network, and emotional health. These reflect many of the same SDoHs raised by our participants and require attention by HCPs caring for patients who experience hypoglycemia. However, HCPs must go beyond simply being knowledgeable about SDoHs but must also be nimble and creative in ways to ameliorate the SDoH effect on hypoglycemia. It may be important to expand members of the team caring for people with diabetes, such as social workers, who can access community resources and provide emotional support (Rabovsky et al., 2017). Our participants identified specific populations such as the vulnerable elderly who may require increased family involvement to lessen their risk of hypoglycemia, parallel to previous research emphasizing the importance of family support (Gunn et al., 2012).

Limitations

This study was conducted in Canada, which has a universal health care system, limiting the transferability of the findings to other countries. Another limitation includes the fact that we did not include patient data. As such, we are unclear as to the type of diabetes (T1DM or T2DM) and specific characteristics of the patients for whom the interviewed HCPs cared. Again, by including a broad group of HCPs, we hoped to mitigate this limitation. Finally, patients' perceptions of the impact of SDoHs on hypoglycemia management require future study.

Conclusion

Using a qualitative approach, this study revealed that HCPs appreciate the impact of the SDoHs on their patients' abilities to manage not only diabetes but also the potential adverse event of diabetes management: hypoglycemia. The two key SDoH themes highlighted by participants as potentially affecting hypoglycemia management (socioeconomic and psychosocial issues) emphasize the need to incorporate a comprehensive social history during each diabetes clinical assessment, especially for patients at risk of hypoglycemia. A call to action is required for HCPs to have access to adequate resources to address the SDoH issues with their most vulnerable patients.

References

- Akhter, K., Turnbull, T., & Simmons, D. (2016). Influences of social issues on type 1 diabetes self-management: Are we doing enough? *Practical Diabetes*, 33, 307–312. <http://dx.doi.org/10.1002/pdi.2061>
- Alexander, G. C., Casalino, L. P., Tseng, C.-W., McFadden, D., & Meltzer, D. O. (2004). Barriers to patient-physician communication about out-of-pocket costs. *Journal of General Internal Medicine*, 19, 856–860. <http://dx.doi.org/10.1111/j.1525-1497.2004.30249.x>
- Ali, S., Stone, M. A., Peters, J. L., Davies, M. J., & Khunti, K. (2006). The prevalence of co-morbid depression in adults with Type 2 diabetes: A systematic review and meta-analysis. *Diabetic Medicine*, 23, 1165–1173. <http://dx.doi.org/10.1111/j.1464-5491.2006.01943.x>
- Barnard, K. D., Skinner, T. C., & Peveler, R. (2006). The prevalence of co-morbid depression in adults with Type 1 diabetes: Systematic literature review. *Diabetic Medicine*, 23, 445–448. <http://dx.doi.org/10.1111/j.1464-5491.2006.01814.x>
- Behforouz, H. L., Drain, P. K., & Rhatigan, J. J. (2014). Rethinking the social history. *New England Journal of Medicine*, 371, 1277–1279. <http://dx.doi.org/10.1056/NEJMp1404846>
- Berkowitz, S. A., Meigs, J. B., DeWalt, D., Seligman, H. K., Barnard, L. S., Bright, O.-J. M., . . . Wexler, D. J. (2015). Material need insecurities, control of diabetes mellitus, and use of health care resources: Results of the Measuring Economic Insecurity in Diabetes study. *Journal of the American Medical Association Internal Medicine*, 175, 257–265. <http://dx.doi.org/10.1001/jamainternmed.2014.6888>
- Booth, G. L., Polsky, J. Y., Gozdyra, P., Cauch-Dudek, K., Kiran, T., Shah, B. R., . . . Glazier, R. H. (2012). *Regional measures of diabetes burden in Ontario*. Toronto, ON, Canada: Institute for Clinical Evaluative Sciences (ICES).
- Borkan, J. (1999). Immersion/crystallization. In B. F. Crabtree & W. L. Miller (Eds.), *Doing qualitative research* (2nd ed., pp. 179–194). Thousand Oaks, CA: Sage.
- Borschuk, A. P., & Everhart, R. S. (2015). Health disparities among youth with type 1 diabetes: A systematic review of the current literature. *Families, Systems, & Health*, 33, 297–313. <http://dx.doi.org/10.1037/fsh0000134>
- Brown, A. F., Ettner, S. L., Piette, J., Weinberger, M., Gregg, E., Shapiro, M. F., . . . Beckles, G. L. (2004). Socioeconomic position and health among persons with diabetes mellitus: A conceptual framework and review of the literature. *Epidemiologic Reviews*, 26, 63–77. <http://dx.doi.org/10.1093/epirev/mxh002>
- Brown, J. B., Lent, B., Stirling, A., Takhar, J., & Bishop, J. (2002). Caring for seriously mentally ill patients. Qualitative study of family physicians' experiences. *Canadian Family Physician Medecin de Famille Canadien*, 48, 915–920.
- Brown, J. B., Ryan, B. L., & Thorpe, C. (2016). Processes of patient-centred care in Family Health Teams: A qualitative study. *CMAJ Open*, 4, E271–E276. <http://dx.doi.org/10.9778/cmajo.20150128>
- Cheng, A. Y., & the Canadian Diabetes Association Clinical Practice Guidelines Expert Committee. (2013). Canadian Diabetes Association 2013 clinical practice guidelines for the prevention and management of diabetes in Canada: Introduction. *Canadian Journal of Diabetes*, 37(Suppl. 1), S1–S3. <http://dx.doi.org/10.1016/j.cjcd.2013.01.009>
- Chiu, C.-J., Wray, L. A., Beverly, E. A., & Dominic, O. G. (2010). The role of health behaviors in mediating the relationship between depressive symptoms and glycemic control in type 2 diabetes: A structural equation modeling approach. *Social Psychiatry and Psychiatric Epidemiology*, 45, 67–76. <http://dx.doi.org/10.1007/s00127-009-0043-3>
- Clark, M. L., & Utz, S. W. (2014). Social determinants of type 2 diabetes and health in the United States. *World Journal of Diabetes*, 5, 296–304. <http://dx.doi.org/10.4239/wjd.v5.i3.296>
- Connolly, V., Unwin, N., Sherriff, P., Bilous, R., & Kelly, W. (2000). Diabetes prevalence and socioeconomic status: A population based study showing increased prevalence of type 2 diabetes mellitus in deprived areas. *Journal of Epidemiology and Community Health*, 54, 173–177. <http://dx.doi.org/10.1136/jech.54.3.173>

- de Groot, M., Anderson, R., Freedland, K. E., Clouse, R. E., & Lustman, P. J. (2001). Association of depression and diabetes complications: A meta-analysis. *Psychosomatic Medicine*, 63, 619–630. <http://dx.doi.org/10.1097/00006842-200107000-00015>
- DiMatteo, M. R. (2004). Social support and patient adherence to medical treatment: A meta-analysis. *Health Psychology*, 23, 207–218. <http://dx.doi.org/10.1037/0278-6133.23.2.207>
- Dowell, A., Stubbe, M., Macdonald, L., Tester, R., Gray, L., Vernall, S., . . . Dew, K. (2018). A longitudinal study of interactions between health professionals and people with newly diagnosed diabetes. *Annals of Family Medicine*, 16, 37–44. <http://dx.doi.org/10.1370/afm.2144>
- Evans, J. M., Newton, R. W., Ruta, D. A., MacDonal, T. M., & Morris, A. D. (2000). Socio-economic status, obesity and prevalence of Type 1 and Type 2 diabetes mellitus. *Diabetic Medicine*, 17, 478–480. <http://dx.doi.org/10.1046/j.1464-5491.2000.00309.x>
- Fisher, L., Skaff, M. M., Mullan, J. T., Arian, P., Glasgow, R., & Masharani, U. (2008). A longitudinal study of affective and anxiety disorders, depressive affect and diabetes distress in adults with Type 2 diabetes. *Diabetic Medicine*, 25, 1096–1101. <http://dx.doi.org/10.1111/j.1464-5491.2008.02533.x>
- Goodall, T. A., & Halford, W. K. (1991). Self-management of diabetes mellitus: A critical review. *Health Psychology*, 10, 1–8. <http://dx.doi.org/10.1037/0278-6133.10.1.1>
- Gunn, K. L., Seers, K., Posner, N., & Coates, V. (2012). ‘Somebody there to watch over you’: The role of the family in everyday and emergency diabetes care. *Health & Social Care in the Community*, 20, 591–598. <http://dx.doi.org/10.1111/j.1365-2524.2012.01073.x>
- Harris, S., Reichert, S., Ryan, B. L., Mequanint, S., Webster-Bogaert, S., Ratzki-Leewing, A., & Brown, J. B. (2016). A population-based study on incidence and associated risk factors for hypoglycemia in Canada: The InHYPO-DM Study. *Canadian Journal of Diabetes*, 40, S11–S12. <http://dx.doi.org/10.1016/j.cjcd.2016.08.033>
- Hempler, N. F., Joensen, L. E., & Willaing, I. (2016). Relationship between social network, social support and health behaviour in people with Type 1 and Type 2 diabetes: Cross-sectional studies. *BMC Public Health*, 16, 198. <http://dx.doi.org/10.1186/s12889-016-2819-1>
- Hill, J., Nielsen, M., & Fox, M. H. (2013). Understanding the social factors that contribute to diabetes: A means to informing health care and social policies for the chronically ill. *The Permanente Journal*, 17, 67–72. <http://dx.doi.org/10.7812/TPP/12-099>
- Holt, R. I. G., Nicolucci, A., Kovacs Burns, K., Escalante, M., Forbes, A., Hermanns, N., . . . the DAWN2 Study Group. (2013). Diabetes Attitudes, Wishes and Needs second study (DAWN2™): Cross-national comparisons on barriers and resources for optimal care—Healthcare professional perspective. *Diabetic Medicine*, 30, 789–798. <http://dx.doi.org/10.1111/dme.12242>
- Holt, R. I. G., Nicolucci, A., Kovacs Burns, K., Lucisano, G., Skovlund, S. E., Forbes, A., . . . Peyrot, M. (2016). Correlates of psychological care strategies for people with diabetes in the second Diabetes Attitudes, Wishes and Needs (DAWN2™) study. *Diabetic Medicine*, 33, 1174–1183. <http://dx.doi.org/10.1111/dme.13109>
- Joensen, L. E., Almdal, T. P., & Willaing, I. (2013). Type 1 diabetes and living without a partner: Psychological and social aspects, self-management behaviour, and glycaemic control. *Diabetes Research and Clinical Practice*, 101, 278–285. <http://dx.doi.org/10.1016/j.diabres.2013.07.001>
- Kelly, T. N., Bazzano, L. A., Fonseca, V. A., Thethi, T. K., Reynolds, K., & He, J. (2009). Systematic review: Glucose control and cardiovascular disease in Type 2 diabetes. *Annals of Internal Medicine*, 151, 394–403. <http://dx.doi.org/10.7326/0003-4819-151-6-200909150-00137>
- Khanolkar, A. R., Amin, R., Taylor-Robinson, D., Viner, R. M., Warner, J. T., & Stephenson, T. (2016). Young people with Type 1 diabetes of non-White ethnicity and lower socio-economic status have poorer glycaemic control in England and Wales. *Diabetic Medicine*, 33, 1508–1515. <http://dx.doi.org/10.1111/dme.13079>
- Kumari, M., Head, J., & Marmot, M. (2004). Prospective study of social and other risk factors for incidence of Type 2 diabetes in the Whitehall II study. *Archives of Internal Medicine*, 164, 1873–1880. <http://dx.doi.org/10.1001/archinte.164.17.1873>
- Leese, G. P., Wang, J., Broomhall, J., Kelly, P., Marsden, A., Morrison, W., . . . the DARTS/MEMO Collaboration. (2003). Frequency of severe hypoglycemia requiring emergency treatment in Type 1 and Type 2 diabetes: A population-based study of health service resource use. *Diabetes Care*, 26, 1176–1180. <http://dx.doi.org/10.2337/diacare.26.4.1176>
- Leiter, L. A., Yale, J.-F., Chiasson, J.-L., Harris, S., Kleinstiver, P., & Sauriol, L. (2005). Assessment of the impact of fear of hypoglycemic episodes on glycemic and hypoglycemia management. *Canadian Journal of Diabetes*, 29, 186–192.
- Liburd, L. C., Jack, L., Jr., Williams, S., & Tucker, P. (2005). Intervening on the social determinants of cardiovascular disease and diabetes. *American Journal of Preventive Medicine*, 29(Suppl. 1), 18–

24. <http://dx.doi.org/10.1016/j.amepre.2005.07.013>
- Maddigan, S. L., Feeny, D. H., Majumdar, S. R., Farris, K. B., & Johnson, J. A. (2006). Understanding the determinants of health for people with Type 2 diabetes. *American Journal of Public Health*, 96, 1649–1655. <http://dx.doi.org/10.2105/AJPH.2005.067728>
- Marmot, M. G., Stansfeld, S., Patel, C., North, F., Head, J., . . . Smith, G. D. (1991). Health inequalities among British civil servants: The Whitehall II study. *Lancet*, 337, 1387–1393. [http://dx.doi.org/10.1016/0140-6736\(91\)93068-K](http://dx.doi.org/10.1016/0140-6736(91)93068-K)
- Nicolucci, A., Kovacs Burns, K., Holt, R. I. G., Lucisano, G., Skovlund, S. E., Kokoszka, A., . . . Peyrot, M. (2016). Correlates of psychological outcomes in people with diabetes: Results from the second Diabetes Attitudes, Wishes and Needs (DAWN2™) study. *Diabetic Medicine*, 33, 1194–1203. <http://dx.doi.org/10.1111/dme.13178>
- Peyrot, M., Rubin, R. R., Lauritzen, T., Snoek, F. J., Matthews, D. R., & Skovlund, S. E. (2005). Psychosocial problems and barriers to improved diabetes management: Results of the Cross-National Diabetes Attitudes, Wishes and Needs (DAWN) Study. *Diabetic Medicine*, 22, 1379–1385. <http://dx.doi.org/10.1111/j.1464-5491.2005.01644.x>
- Piette, J. D., Heisler, M., Horne, R., & Caleb Alexander, G. (2006). A conceptually based approach to understanding chronically ill patients' responses to medication cost pressures. *Social Science & Medicine* (1982), 62, 846–857. <http://dx.doi.org/10.1016/j.socscimed.2005.06.045>
- Piette, J. D., Heisler, M., & Wagner, T. H. (2004). Cost-related medication underuse: Do patients with chronic illnesses tell their doctors? *Archives of Internal Medicine*, 164, 1749–1755. <http://dx.doi.org/10.1001/archinte.164.16.1749>
- Piette, J. D., Wagner, T. H., Potter, M. B., & Schillinger, D. (2004). Health insurance status, cost-related medication underuse, and outcomes among diabetes patients in three systems of care. *Medical Care*, 42, 102–109. <http://dx.doi.org/10.1097/01.mlr.0000108742.26446.17>
- Rabovsky, A. J., Rothberg, M. B., Rose, S. L., Brateanu, A., Kou, L., & Misra-Hebert, A. D. (2017). Content and outcomes of social work consultation for patients with diabetes in primary care. *Journal of the American Board of Family Medicine*, 30, 35–43. <http://dx.doi.org/10.3122/jabfm.2017.01.160177>
- Raphael, D., Anstice, S., Raine, K., McGannon, K. R., Kamil Rizvi, S., & Yu, V. (2003). The social determinants of the incidence and management of Type 2 diabetes mellitus: Are we prepared to rethink our questions and redirect our research activities? *Leadership in Health Services*, 16, 10–20. <http://dx.doi.org/10.1108/13660750310486730>
- Sandelowski, M. (2000). Whatever happened to qualitative description? *Research in Nursing & Health*, 23, 334–340. [http://dx.doi.org/10.1002/1098-240X\(200008\)23:4<334::AID-NUR9>3.0.CO;2-G](http://dx.doi.org/10.1002/1098-240X(200008)23:4<334::AID-NUR9>3.0.CO;2-G)
- Sawka, A. M., Boulos, P., Talib, A. S., Gafni, A., Thabane, L., Papaioannou, A., . . . Gerstein, H. C. (2007). Low Socioeconomic Status and Increased Risk of Severe Hypoglycemia in Type 1 Diabetes: A Systematic Literature Review. *Canadian Journal of Diabetes*, 31, 233–241. [http://dx.doi.org/10.1016/S1499-2671\(07\)13010-0](http://dx.doi.org/10.1016/S1499-2671(07)13010-0)
- Seligman, H. K., Davis, T. C., Schillinger, D., & Wolf, M. S. (2010). Food insecurity is associated with hypoglycemia and poor diabetes self-management in a low-income sample with diabetes. *Journal of Health Care for the Poor and Underserved*, 21, 1227–1233.
- Smalls, B. L., Gregory, C. M., Zoller, J. S., & Egede, L. E. (2015a). Assessing the relationship between neighborhood factors and diabetes related health outcomes and self-care behaviors. *BMC Health Services Research*, 15. Retrieved from <http://search.proquest.com/docview/1779685691/abstract/A0A27F7D50B2430DPQ/1>
- Smalls, B. L., Gregory, C. M., Zoller, J. S., & Egede, L. E. (2015b). Direct and indirect effects of neighborhood factors and self-care on glycemic control in adults with Type 2 diabetes. *Journal of Diabetes and Its Complications*, 29, 186–191. <http://dx.doi.org/10.1016/j.jdiacomp.2014.10.008>
- Stewart, M., Weston, W., Brown, J. B., McWhinney, I., McWilliam, C., & Freeman, T. (2014). *Patient-centered medicine: Transforming the clinical method* (3rd ed.). London, UK: Radcliffe.
- Walker, R. J., Gebregziabher, M., Martin-Harris, B., & Egede, L. E. (2014a). Independent effects of socioeconomic and psychological social determinants of health on self-care and outcomes in Type 2 diabetes. *General Hospital Psychiatry*, 36, 662–668. <http://dx.doi.org/10.1016/j.genhosppsych.2014.06.011>
- Walker, R. J., Gebregziabher, M., Martin-Harris, B., & Egede, L. E. (2014b). Relationship between social determinants of health and processes and outcomes in adults with type 2 diabetes: Validation of a conceptual framework. *BMC Endocrine Disorders*, 14, 82. <http://dx.doi.org/10.1186/1472-6823-14-82>
- Walker, R. J., Smalls, B. L., Campbell, J. A., Strom Williams, J. L., & Egede, L. E. (2014). Impact of social determinants of health on outcomes for Type 2 diabetes: A systematic review. *Endocrine*, 47, 29–48. <http://dx.doi.org/10.1007/s12020-014-0195-0>
- Walker, R. J., Smalls, B. L., & Egede, L. E. (2015). Social determinants of health in adults with type 2 diabetes—Contribution of mutable and immutable

- factors. *Diabetes Research and Clinical Practice*, 110, 193–201. <http://dx.doi.org/10.1016/j.diabetes.2015.09.007>
- Williamson, C., Glauser, T. A., Burton, B. S., Schneider, D., Dubois, A. M., & Patel, D. (2014). Health care provider management of patients with type 2 diabetes mellitus: Analysis of trends in attitudes and practices. *Postgraduate Medicine*, 126, 145–160. <http://dx.doi.org/10.3810/pgm.2014.05.2764>
- Zhang, X., Norris, S. L., Gregg, E. W., & Beckles, G. (2007). Social support and mortality among older persons with diabetes. *The Diabetes Educator*, 33, 273–281. <http://dx.doi.org/10.1177/0145721707299265>
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